

Liquidity and Macroeconomics: a vision and my research agenda

Saki Bigio

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This is an unconventional research statement. Here, I talk not only about the content of papers, but also about the context that motivated each project. Developing a research agenda is an act of exploration: as scientists, we are on the lookout, but we do not know where research will take us. I think others, especially students, may find knowing about the process as useful as the finished results. I therefore organize the essay around the policy questions, gaps in the literature, technical ideas, and conversations that connected one project to the next.

Along the way I offer some lessons about what has—and has not—worked for me. The sidenotes contain mostly thoughts about economics, following the ancient scholarly tradition of my people. My tastes will be evident.

Ok, let's get started. I am a macroeconomic theorist. My goal in life is to build quantitative models to understand the world. I say quantitative because I see a value not only in figuring out forces behind mechanisms, but also whether they are meaningful.¹ My research lies at the intersection of business cycles, money and banking, and corporate finance. It asks how financial architectures and their imperfections shape macroeconomic performance, and how policy can use those imperfections to improve outcomes.

¹ I see a value in disciplining the importance of a mechanism. Matching many variables, is not an end in itself, unless someone is going to make use of a large model. The challenge is to identify amplification that is intrinsic to a mechanism.

How did I kick-start my agenda?

My research agenda began during my third year at NYU, amid the 2008–2009 financial crisis. Most macroeconomists will acknowledge that, at the time, business cycles were studied as phenomena independent of capital market imperfections. Macroeconomics faced widespread criticisms from within and outside economics. In fact, in its last issue of 2013, five years after the Lehman Brothers debacle, *The Economist* put it in the following terms:

“Mainstream macro models fail to represent some of the most basic realities of the financial system. One reason is that doing so is hard. Another is that for a long time it did not seem to make much difference. In the absence of crises, the activities of the financial sector can appear irrelevant for long stretches of time. Small wonder so many academics model the economy as if banks and other intermediaries simply do not exist. The crisis, which was completely unanticipated by the vast majority of academics and policymakers, revealed some of the drawbacks of these shortcuts.”
(*The Economist*, 2013)

There was much truth in that comment—there still is. For my part, the weekend of the fall of Bear Stearns, March 2008, I recall discussing its with some Wall Street people *I still think it is a good exercise for a macro student to translate Wall Street narratives into modeling language.*² The conversation was about what the government should do to restore liquidity. I was embarrassed that, as a PhD

² Often, these narratives are logically inconsistent. Often, they are on to something or teach you about how people think.

student, I couldn't even define market liquidity!³ Macroeconomics, of course, reacted by quickly developing models where financial market imperfections are at the epicenter. Today, dedicated societies, research programs, and NBER working groups, and has become influential in policy.

In my case, I was fortunate that Mark Gertler taught his financial accelerator model (Bernanke and Gertler, 1989; Bernanke, Gertler and Gilchrist, 1996) during the last module of a course entirely dedicated to the new-Keynesian framework. I had no background on corporate finance or asset pricing. I knew very little about this, but I thought it was an important area. After the last lecture, during the summer of 2008, I began surveying some of the few existing models with financial frictions—the sequence of papers by Allen and Gale (1998, 2000), Holmstrom and Tirole (1997, 1998), Kiyotaki and Moore (1997), Bernanke and Gertler (1989) and Caballero and Krishnamurthy (2001) and some others.⁴ *Second-year students often ask how to begin a research program. My answer is always the same: spend the summer writing a critical literature review.*⁵ I think I identified a void thanks to that exercise.

What causes contractions in liquidity? Can contractions in liquidity quantitatively explain business cycles?

Papers: Bigio (2015, AER); Bigio and Schneider (2017, EER).

I began working on Bigio (2015) during my third year, the fall of 2008. I was motivated by the idea that the 2008-2009 financial crisis began with an abrupt collapse in the funding of many businesses. The papers I surveyed had focused on limited enforcement or moral hazard in the development of investment projects. Yet, when we think of operating firms, it seems evident that most of the capital stock is worth many multiples of their investment needs. For that reason, I began to investigate frictions that could prevent firms from raising funds despite having capital worth multiple times their needs—frictions that would endogenously reduce the liquidity of their capital.

I was lucky to be invited by Roberto Chang, a dear friend and mentor, to see a presentation by Nobu Kiyotaki at Rutgers that summer—Kiyotaki and Moore (2008, a paper whose final version met the printing press two decades after its first draft). I loved the paper and thought it would be a good building block. In Kiyotaki and Moore, limited enforcement prevents agents from transacting solely with promises. Therefore, firms sell their past capital stock to finance new investment. However, in Kiyotaki and Moore, for the aforementioned reason, only an exogenous share of the capital stock can be sold. But why? At the time, the view among practitioners was that the financial crisis stemmed from asymmetric information about valuing collateral. It occurred to me that recessions could be periods where the heterogeneous returns to existing capital fan out, thereby aggravating lemons problems in recessions. After all, recessions are periods with a lot of churning. The question—what causes contractions in liquidity—was so natural that a similar model was developed independently as a job market paper (Kurlat, 2013) as I was finishing my fourth year.

In Bigio (2015), each firm operates a continuum of capital units whose returns are private information. Each unit receives a random depreciation shock whose

³ It is a problem that we don't have a proper mathematical definition of liquidity. Here is mine: Liquidity as a property of an asset. An asset is said to be liquid if gains from trade are sufficient conditions for trade.

⁴ Actually, the first thing I did after Mark's final class was to buy Jean Tirole's Corporate Finance Textbook. I think most people don't read textbooks anymore, but they are great: they provide unified frameworks, cover the literature and motivations, and provide exercises which you can try solving mentally. What better resource could there be?

⁵ Emphasis on *critical*.

magnitude is the source of private information. There are two technical “tricks” in that paper. First, the continuum assumption is convenient because one can derive marginal conditions; second, shocks to units are permanent, which makes aggregation possible.⁶

In [Bigio \(2015\)](#), I encountered a formula that determines the marginal asset sold under asymmetric information, for asset qualities $\lambda \sim F$. The marginal asset sold λ^* is given by the following marginal condition:

$$\underbrace{\frac{\lambda^*}{\mathbb{E}_F[\lambda|\lambda < \lambda^*]}}_{\text{lemons premium}} = \underbrace{(1 + r(\mathbb{E}_F[\lambda|\lambda < \lambda^*]))}_{\text{marginal value of liquidity}}$$

The lemons premium is the ratio of the internal valuation of the marginal asset relative to the average valuation in the pool of sold assets. In equilibrium, the premium must equal the marginal value of relaxing financial frictions with a marginal unit of funding— r encodes the value of relaxing constraints as in [Kiyotaki and Moore](#). From that marginal unit, you get the total funding: $x = \lambda^* \mathbb{E}_F[\lambda|\lambda < \lambda^*]$.

Naturally, the balance tilts towards less liquidity when the asset-quality distribution becomes more dispersed, providing a theory of time-varying liquidity. In a business cycle model, a contraction in liquidity translates into less production, which can occur even if the production possibility frontier is untouched.

As I was working through the theory in [Bigio \(2015\)](#), my co-advisor, Ricardo Lagos, suggested I also figure out whether the liquidity shocks in [Kiyotaki and Moore](#) could have a quantitative bite. The paper split into a separate thesis chapter that eventually became a paper co-written with a UCLA graduate student, Andres Schneider. [Bigio and Schneider \(2017\)](#) makes a surprising but simple point about models with contractions in liquidity: regardless of the source of the contraction, without additional strong amplification mechanisms—sticky prices or strategic complementarities—models with financial frictions that limit investment simply cannot account for the observed magnitudes of the Great Recession.⁷ The reason is that investment is such a small share of the capital stock that even if we shut down investment entirely for a year, the displacement of the production possibility frontier is too small to account for significant recessions.

It became evident that the macro-finance agenda was directly importing off-the-shelf models from corporate finance but was having difficulty producing quantitatively significant business-cycle fluctuations. If you write about corporate finance, you probably want to talk about financing large investment projects, where a bad financial decision may derail a firm. But when you talk about business cycles, you need something else to deliver extra amplification. I was reluctant to add an amplification mechanism—like sticky wages—because such mechanisms amplify all shocks, including the policy interventions that could have corrected the financial ones. Why should macro-finance develop into an independent subfield if financial shocks are like other shocks? Learning from the preliminary work of [Bigio and Schneider \(2017\)](#), I decided to try adding funding needs to hiring decisions to [Bigio \(2015\)](#). Changing the location of frictions produces strong output responses, even from mild increases in capital-quality dispersion. This allowed the paper to replicate the output and investment pattern observed during the Great Recession.

⁶ Aggregation is an underappreciated art: finding assumptions under which models aggregate is an important skill. Even if unrealistic, the models that admit aggregation produce useful benchmarks. I learned the value of this skill by studying [Angeletos \(2007\)](#), a gem.

⁷ [Bigio and Schneider \(2017\)](#) also argues that the asset pricing implication of such models was counterfactual: a scarcity logic applies in those models and, thus, the price of capital increases when liquidity limits the amount of investment. Both points are shared in common with contemporary work by [Shi \(2015\)](#).

Why are recoveries from financial crises so slow, and why do banks struggle to raise equity during crises?

Papers: [Bigio and d'Avernas \(2021, AEJ: Macro\)](#).

By 2011, it was evident that the Great Recession was different from other recessions. The recovery was taking forever. Emerging evidence was beginning to show that financial crises cause deeper recessions and slower recoveries (see [Cerra and Saxena, 2008](#); [Reinhart and Rogoff, 2009](#)). Theoretically, we relate these crises to the evolution of bank equity. I recall listening to an interview with the Chairman of the Federal Reserve, Ben Bernanke. When asked when the crisis would end, he answered, “When banks start raising capital on their own.”⁸

⁸ “The Chairman,” 60 Minutes, CBS News, March 15, 2009.

In the aftermath of the Great Recession, several papers had already incorporated intermediaries into business cycle models—(see for example [Gertler and Kiyotaki, 2010](#); [He, Khang and Krishnamurthy, 2010](#); [Gertler and Karadi, 2011](#); [Brunnermeier and Sannikov, 2014](#)). However, in all of those models, the aftermaths of crises are periods of high excess returns for banks because a logic of scarcity applies—the scarce resource, bank equity, produces high returns. In my view, this still represents a problem for canonical models: without a counterforce, these models imply that banking crises should be times when banks raise capital faster and extraordinary returns should lead to fast a recovery of their equity. Bernanke’s answer precisely pokes his finger at what models were missing.

Furthermore, I found it very unappealing that these models contain nothing intrinsically special about intermediaries: they are simply relabelings of earlier models borrowed from corporate finance. Actual banks are distinctive for several reasons—more on this below.

By my fifth year, [Bigio \(2015\)](#) was already under revision, but the area was becoming crowded. Tom Sargent, one of my advisers, suggested I could use another year to start my first job with one more project. So I decided to address the challenge of writing a model where bank returns are low and there is something special about bank intermediation. I wanted a theory in which severe financial losses lead to slow recoveries, while mild ones are only temporary. I embraced—and still hold—the view in [Gorton \(2012\)](#) that the central role of banks is to create securities that are immune to adverse selection—i.e., to issue money. That draft eventually became a much more elegant paper [Bigio and d'Avernas \(2021\)](#) after being co-authored with my UCLA student, Adrien d'Avernas.

We illustrate the underlying mechanism with some graphs which I replicate here. As with any intermediary, a financial intermediary is an agent that buys from the supply schedule and sells to the demand schedule. Figure 1 illustrates the mechanism. A financial intermediary buys along the supply schedule and sells along the demand schedule.

In the two panels of Figure 1, the two curves represent aggregate demand and supply schedules $D(Q)$ and $S(Q)$, respectively, given an intermediated quantity Q . The intermediaries’ marginal profit is simply $\Pi(Q) = S(Q) - D(Q)$. If some friction limits the volume of intermediation, Q , there is a positive arbitrage. In models with financial intermediaries, a multiple of net worth caps Q .

The shapes of the demand and supply schedules govern the behavior of marginal

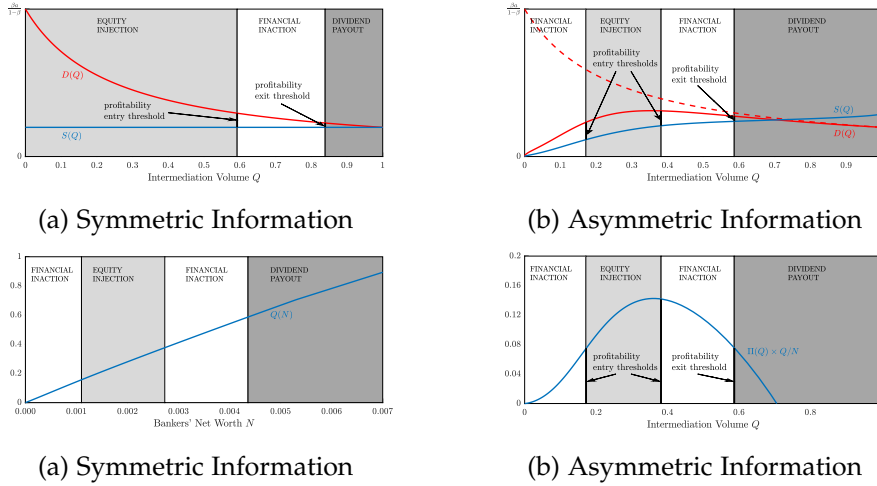


Figure 1: Financial Intermediation w/ and w/o Asymmetric Information

profits. Without information asymmetries, marginal profits decrease in Q —in panel (a) of Figure 1, the gap between the schedules narrows as Q grows—and so does the return to bank equity.

Marginal profits influence the evolution of net worth in two ways: directly, by affecting retained earnings, and indirectly, by attracting outside equity injections. If equity is large and marginal profits are low, banks pay out dividends because the return is low. The opposite occurs whenever net worth is below a threshold, and equity injections, attracted by returns, replenish net worth. Because the entry and exit profit levels are not the same, there is an intermediate equity inaction region. Within that region, equity can increase, but only through retained earnings. If shocks reduce bank net worth, this logic applies, and equity injections and retained earnings operate as a stabilizing mechanism. Asymmetric information alters this stabilizing force. When intermediaries purchase capital under asymmetric information, both the quantity and the quality of assets increase with the purchase price, and the effective demand is backward-bending. A direct consequence of this backward-bending demand is that marginal profits are no longer decreasing: they are potentially hump-shaped, as in panel (b) of Figure 2. The hump generates two inaction regions in the equity-injection policy—shown in panel (a)—instead of a single one.

The inaction regions impair the stability of financial intermediation. A shock that produces small equity losses that only send the economy to the injection region is stabilized when information is symmetric. If losses are significant, the economy enters the leftmost inaction region, and the economy loses its tendency to return to equilibrium.⁹

Bigio and d’Avernas embeds this logic into a fully fledged general equilibrium model where the size of shocks matter. Large shocks capsize the economy, but small ones are stabilized.

...But firms don’t sell capital; they just borrow against it...

Papers: Bigio and Shi (2025, Restud).

An unappealing feature of Bigio (2015) and Bigio and d’Avernas (2021), and

Figure 2: equity Injection Decisions / Marginal Profits

⁹ Years later I learned that Axel Leijonhufvud had conjectured precisely this—his “corridor hypothesis”: the economy self-stabilizes within a range, and loses that homeostasis outside it—decades earlier, at the department where I would eventually land. The questions that feel newest are often the oldest ones, waiting for tools.

related models, is that capital is sold in those models. In practice, firms *pledge* assets as collateral. To be honest, I heard this criticism in a referee report—that referee’s question, and the question stayed with me: what if firms pledge collateral rather than sell it (under private information when)? The problem with a lemons market is that the seller of a good car loses the difference between the value of his car and the average value of cars. In used car markets—a car seller has no reason to repurchase the car. A firm, by contrast, still needs its capital after raising funds. This makes a contract with a repurchase option—a repo—a natural response to adverse selection.

I began working on this question but I didn’t know how to think about equilibrium contracting. I started talking to Liyan Shi around 2017. Her UCLA job-market paper studied equilibrium contracts under non-compete clauses. We decided to think together about a market for repurchase contracts in a setting with private information. That collaboration became [Bigio and Shi \(2025\)](#).

In [Bigio and Shi \(2025\)](#), we argue that in asset markets with asymmetric information, pledging collateral, which is akin to signing repos, is a natural resolution to adverse selection. We consider a market where borrowers require funds but have private information about existing assets, and lenders compete by offering repurchase options (repos). Repos resolve the adverse-selection problem in the sense that all assets are used and no asset remains idle. Yet, the value of assets is not fully extracted as liquid funds and, moreover, competition induces inefficient cream-skimming relative to a planner. Several equilibrium concepts studied in the paper give you slightly different formulas, but the essence is the same in all cases.

The equilibrium contract yields a closed-form expression for liquid funding, x , using the same objects as [Bigio \(2015\)](#):

$$x = \mathbb{E}_f \left[\min \left\{ \lambda, f^{-1} \left(\frac{r(x)}{1 + r(x)} \right) \right\} \right].$$

Importantly, the contract also delivers a default rate and an equilibrium interest rate. I remember the afternoon Liyan arrived with the idea that our equilibrium could be characterized as a maximization problem. I asked her: why don’t we take the first-order condition? We did it in two minutes and that step produced the formula above and, I believe, makes the theory portable to much richer settings.

How do contractions in liquidity get amplified by production networks?

Papers: [Bigio and La’O \(2020, QJE\)](#).

I met Jennifer La’O, known amongst friends as Jen-Jen, during her job-market talk at NYU in 2010. She had written an influential paper on sentiment shocks and was presenting work at the intersection of noisy information and financial frictions. I sent her a few comments and we began exchanging emails. *I guess it pays off not to be shy in giving feedback to people you admire.*

I had a view based on what I had learned in [Bigio \(2015\)](#). We both agreed that existing models of financial frictions, which focused on the investment margin, had difficulty generating business-cycle fluctuations. In turn, we also knew that labor rigidity from working-capital constraints could generate large fluctuations, but required excessively high financing premia. So that was another problem.

Some papers were trying to model capital misallocation, exporting the [Hsieh and Klenow \(2009\)](#) logic to business-cycle frequency, but seemed at odds with capital reallocation data—and requires big differences in profitability to work.¹⁰

Jen-Jen and I wanted to respond to these challenges. We began thinking of a framework where the sum of small frictions is what really matters—the idea that double, triple, quadruple marginalization can escalate. So we began thinking about inter-firm linkages in 2011. Our thought was that with firm linkages, financial shocks that only affect a subsector could spill over to other sectors through supply and demand channels. Small contractions in liquidity and modest credit spreads could then cumulate into large aggregate quantity responses. This could also offer an answer to [Chari and Kehoe](#): the financing of costs, rather than capital investment, may be the right metric for the firms' funding needs.

We had both recently seen the study on the amplification of TFP shocks in production networks [Acemoglu, Carvalho, Ozdaglar and Tahbaz-Salehi \(2012, ECMA\)](#) and began exploring how we could introduce financial frictions, or distortions more generally, due to financial networks into production networks. This project eventually became [Bigio and La'O \(2020\)](#). In [Bigio and La'O](#) we consider a static, multi-sector, general equilibrium model of inter-sectoral trade. We introduce exogenous wedges that distort the input choices of firms.¹¹ The paper is about how sectoral distortions aggregate. We show that distortions manifest in total factor productivity (TFP) and in a labor wedge that does not appear in [Acemoglu et al.](#) Near efficiency—or equivalently, for moderate funding spreads—distortions are of second order for aggregate TFP, but first-order for the labor wedge. The impact of sectoral shocks on the labor wedge is proportional to the sectoral sales relative to GDP—Domar weights. The first part was no surprise because productive efficiency is maximal when distortions are zero. However, distortions drive wedges between sectoral prices and marginal costs, and the effects of these wedges compound. At the aggregate level, the price of the final consumption good is distorted away from its marginal cost of production, and a labor wedge arises. The logic applies to any factor with a vertical supply schedule.

Here's the key equation in this paper. Aggregate quantities can always be represented as a prototype economy, $C = A \cdot L$, where A is an endogenous TFP and Λ is an endogenous aggregate labor wedge that drives a gap between the marginal rate of substitution and the marginal rate of transformation. The central result maps the sectoral distortions φ_i into these two aggregate wedges:

$$\underbrace{\frac{d \log A}{d \log \varphi_i} = 0}_{\text{no first-order TFP loss}} \quad \text{and} \quad \underbrace{\frac{d \log \Lambda}{d \log \varphi_i} = \lambda_i}_{\text{labor wedge moves with the Domar weight}}$$

This is the counterpart of Hulten's theorem for distortions: productive efficiency is maximal at $\varphi = 1$, so misallocation only shows up at second order, but the compounding of wedges along the network shifts the labor wedge one-for-one with sectoral sales over GDP. Output moves because of the labor wedge.

That paper ends with a quantitative exercise motivated by our original thoughts. We feed the model with sectoral-level credit spreads obtained from [Gilchrist and Zakrajšek \(2012\)](#) and demonstrate that financial distortions can indeed explain a large portion of decline in output of the Great Recession. For that, the model

¹⁰ [Buera and Moll \(2012\)](#) formalized these points. Some papers went further, (e.g., [Chari and Kehoe, 2009](#)), by challenging the idea that a credit contraction on firms was quantitatively meaningful or that corporations could finance their total capital expenditures with retained earnings (see also [Zetlin-Jones and Shourideh, 2012](#)).

¹¹ These distortions may be also rationalized as the result of marginal tax rates or monopolistic markups, so the final version took that generality although our original motivation was financial.

needs a labor-supply elasticity close to those used in the business-cycle literature.

In retrospect, a limitation of [Bigio and La'O](#) is that we work with intermediate-input elasticities of substitution equal to 1, which gives economies too much flexibility—at least at business-cycle frequency—making the labor wedge the primary transmission mechanism. In some sense, the business-cycle amplification is not that satisfactory because the standard labor-supply elasticity used in the business-cycle literature is a stand-in for other frictions.

Relative to our original motivation, the paper was a useful starting point. Later work—particularly [Baqee and Farhi \(2019\)](#)—is more interesting because lower intermediate-input elasticities make the mechanism considerably more powerful. I have a solution to a version of [Bigio and La'O](#) with non-unit elasticities written on the flipside of a book. In retrospect, that version would have allowed us to make additional statements about flexibility. *The lesson is not to abandon an interesting extension merely because the original paper is finished; good ideas often have a long life.* I am glad David and Emmanuel pushed their results.

What are payment-chain crises?

Papers: [Bigio, Mendez and Van Patten \(2025a, draft\)](#).

2008 defined my research agenda, but 1998 steered me toward economics. That was the year of the Russian crisis which spilled over to my home country, Peru. It was personally traumatic. My mother's company was struggling as loans were being called. The company could not repay its suppliers and creditors because its customers could not pay their bills; one of those customers was the government itself. Latin Americans call these episodes “La crisis de la cadena de pagos” or “payment-chain crises”—a scenario that still freaks policymakers out.

[Bigio and La'O](#) is an attempt to model relations among financially constrained firms, but time—wasting time, rather—plays no role. Instead, payment-chain crises instead evoke supply chains in which firms must wait to be paid before they can pay others. While they wait, resources sit idle and utilization falls.

I raised this observation with Diana Van Patten, also a former UCLA student and now at Yale. Together with Esteban Mendez of the Central Bank of Costa Rica, she connects ideas from trade to transaction-level data. In [Bigio et al. \(2025a\)](#), our goal is to formalize payment-chain crises and test whether payment delays slow the activity of firms along the supply chain.

This is the key equation in that paper. There's a network of transactions in the economy—initiated by different firms, let's say. Some transactions are funded, some are not. Those that are, depend on the execution of earlier transactions. The key assumption is that transactions cannot be executed until funded, and delays in execution translate into delays in production. If the share of transactions that must wait to be funded is μ , then the economy's TFP is $\mathcal{A}(\mu)$. The mapping of funded transactions to TFP, in fact, depends on how the network is organized:

$$\underbrace{\mathcal{A}(\mu) = \delta \frac{1-\mu}{1-\delta\mu}}_{\text{random assignment}} \quad \text{versus} \quad \underbrace{\mathcal{A}(\mu) = \delta \frac{1 - [p\delta^u + (1-p)\delta^{\bar{n}}]}{1-\delta} \frac{1-\mu}{\mu}}_{\text{efficient assignment}},$$

where δ represents the idleness of resources as transactions take time to clear. As μ

risers, $A(\mu)$ falls at an increasing rate: the more orders must wait to be paid before they can pay, the longer the chain of payments, the further the queue is, and the more output is idled away waiting. Payment chains can operate fluidly, but they can also seize up if firms are sufficiently constrained.

We find evidence of this force. Costa Rican data allows to combine the universe of electronic payment receipts—which time-stamp every interbank transfer between firms—with VAT invoices. Two facts stand out. First, many firms make payments conditional on receiving payments. Moreover, we observe that many goods are shipped only when payments are made. The finding is strongest for small firms and fades for firms with more assets or more access to loans—exactly the ranking you would expect from the theory. Classifying firms by this behavior, about 61% depend on earlier payments and chains average five links—the firm at the end of one waits between 5 and 25 business days a year to receive a product. We also study a natural experiment: the closure of a financial company cut liquidity lines to many firms, and we observe the effects trickling down the supply chain.

What I like about this theory is that loss comes from the loss of an important production input: time. Nothing is destroyed, no technology regresses, and there is no shortfall of demand: the economy is at full capacity whenever liquidity is abundant. The loss is pure waiting—labor, machines, and half-finished inputs sitting idle while someone upstream waits to be paid—and national accounts book it as a fall in TFP. The policy conclusions, obtained by introducing the mechanism into a business-cycle model, are the part I did not anticipate. Once payment speed is an externality—nobody internalizes that paying late slows the whole chain—business cycles are inefficient. Fiscal policy can then have a positive multiplier, but with a condition that has nothing to do with aggregate demand: the government has to pay upfront. Public spending that is itself chained lengthens the queue and does damage. I find this a rather satisfying reinterpretation of what a fiscal multiplier is—not about demand, but about unclogging payments.

How does monetary policy affect bank liquidity, and why does that matter for the supply of credit?

Papers: Bianchi and Bigio (2022, Econometrica).

Monetary policy also changed during the years of my doctoral dissertation. During 2008, the interbank market froze. Banks responded by slashing new credit as these events unfolded. The world was fortunate to have a financial economist and historian, Ben Bernanke, as chair of the Federal Reserve. I say this because by 2008, the leading theoretical framework for monetary policy analysis, the New Keynesian model, abstracted completely from implementation issues: it had no notion of an interbank market or of its direct influence on credit markets. In theory and practice, the dominant paradigm was to focus policy on a sole instrument, the nominal interest rate and communication about its future path, while forgetting about money or implementation altogether.¹² Bernanke understood the plumbing and had the vision to think beyond what-interest-to-set-next and carried out an unprecedented private-asset purchase program. It was bold because the fiscal and inflationary consequences were uncertain.

¹² The textbook treatment of the New Keynesian model (e.g. Woodford, 2004; Galí, 2015) followed the dictum in Woodford (1998) that monetary policy can be analyzed without money. Even within the New Monetarist framework, scholars were interested in properties of monetary and credit exchange, but not on the impact of central bank policy on credit.

By my graduation in 2012, banks had accumulated large amounts of central bank reserves without a substantial expansion in lending. Not surprisingly, the role of banks in monetary policy transmission was at the center of policy discussions (see [Keister and McAndrews, 2009](#)). There was no modern macroeconomic framework that explicitly modeled the implementation of monetary policy through the banking sector.

That year Javier Bianchi visited NYU. We were talking about the issue and about how central banks implement monetary policy in practice. That is, what does it mean to pick an interest rate? After discussing the issue, we began working on a tractable general equilibrium model that could articulate the credit channel of monetary policy and its implementation through the supply of reserves to the banking system. I had been reading Stigum's *Money Market* ([Stigum and Crescenzi, 2007](#)) to understand the flow of funds in order to calibrate the model in [Bigio and d'Avernas \(2021\)](#).¹³ Working through the T-accounts and the settlements required to sustain payments made clear that liquidity management is essential to understanding how central banks affect interest rates. Liquidity management lies at the heart of [Bianchi and Bigio \(2022\)](#).

There, bank portfolios are composed of deposits, loans, government bonds, and reserves. When a bank grants a loan, it simultaneously issues deposits and gains intermediation profits. However, deposits circulate. When a deposit is moved out of a bank, another bank absorbs that liability. As occurs in practice, that transfer is settled with reserves because loans are illiquid.¹⁴ [Bianchi and Bigio](#) take this architecture as given and treat deposit withdrawals as unpredictable. If a deposit withdrawal is too large, the bank ends short of reserves. The bank can sell bonds in exchange for reserves, but this may not be enough to cover its deficit. At that point, the bank must incur the expense of borrowing reserves, either from the discount window at a penalty rate or from the interbank market. Because the interbank market is frictional, possibly because not all banks trust each other, the probability of finding a counterpart in the interbank market depends on the scarcity of reserves. Thus, the supply of reserves affects the interbank market's terms of trade and, thus, the degree of liquidity risk. By holding a large buffer of liquid assets composed of bonds and reserves, a bank reduces its exposure to liquidity risk at the expense of intermediation profits. Monetary policy shifts the supply of bank credit by tilting this trade-off and, with it, liquidity premia. Here's the relevant equation from [Bianchi and Bigio](#):

$$\mathcal{LP} = [(1 - \Phi(-\mu)) \bar{\chi}^+(\theta) + \Phi(-\mu) \bar{\chi}^-(\theta)].$$

This equation tells you that the liquidity premium $\mathcal{LP}$ —the spread of loans over reserves—depends on the liquidity ratio, μ , that is, the ratio of reserves to deposits—a notion of how well covered deposit outflows are. In turn, θ measures the market tightness in the interbank market: the ratio of aggregate reserve deficit positions to surpluses. The functions $\bar{\chi}^\pm$ dictate the average cost of closing a position for surplus and deficit banks and Φ is the C.D.F. of deposit outflows. You can view these objects as convenience yields and μ as a target the central bank can control to affect outcomes.

Monetary policy moves liquidity premia by steering this ratio—that is the lending channel at work. Of course, in the paper we deal with general equilibrium

¹³ *Stigum's Money Market* should be a reference for anyone studying macro-finance. Working through its T-accounts reveals a great deal about the pursuit of a foundation of why deposits circulate and loans not, in [Bigio, Weill and Zuniga \(2024a\)](#) discussed below. [de Cavalcanti, Erosa and Temzelides \(1999\)](#) captures well why banks settle with reserves and not with an infinite ledger but I think there's more work to be said.

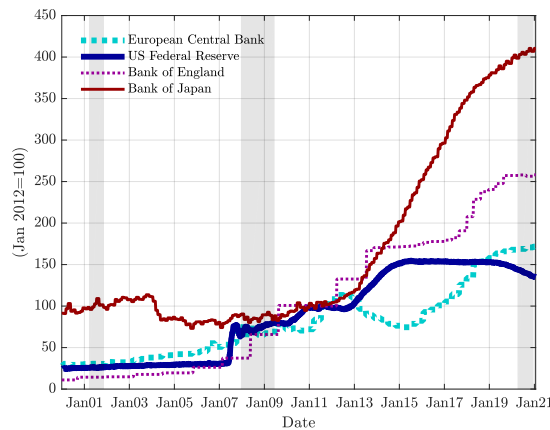


Figure 3: Total Asset Holdings of Major Central Banks

effects, the anticipation of policy, and resolve the details of price determination.

Analyzing monetary policy transmission through the banking system leads to novel insights: In contrast to models in which reserve requirements exogenously determine the demand for reserves, monetary policy here affects the risk-return trade-off between holding reserves vis-à-vis loans. The central bank alters this trade-off through both conventional and unconventional open market operations and by setting interest rates on reserves and discount window lending. Although the composition between government bonds and reserves is irrelevant for an individual bank, the composition matters at the macro level. We show that swaps of bonds for reserves have aggregate effects on liquidity premia by altering the interbank market tightness. Moreover, by absorbing illiquid assets into the central bank's balance sheet, unconventional open market operations have more potent effects than conventional ones.

A central insight of the paper is that the implementation of monetary policy matters: interest on reserves and the central bank's balance sheet constitute independent policy instruments. Crucially, we show how configurations that achieve the same target for the interbank market rate generate a different lending rate and pass-through. In particular, configurations with a larger balance sheet induce a larger credit supply and a higher pass-through from the interbank-market rate to the lending rate. These findings imply that the questions on setting a target for the policy rate and how to implement it must be analyzed together.

An application of that paper is to examine the credit crunch during the U.S. financial crisis after 2008. Back then, it was not clear why banks were holding so many reserves without extending loan expanding lending.¹⁵ In [Bianchi and Bigio](#) we devise a procedure to reverse engineer the shocks needed to discern the importance of these conflicting stories. The key finding is that disruptions in the interbank market played a substantial role around the time of the Lehman Brothers bankruptcy but by 2010, loan demand and borrower creditworthiness had become the dominant factors. I believe this finding calls for a "not-so-obvious theory where an initial credit crunch eventually becomes a demand crisis."¹⁶

¹⁵ I recall: Joseph Stiglitz argued that the significant equity losses of 2008 were to blame, whereas Robert Lucas, argued that the interbank market was jammed. Martin Feldstein wrote an op-ed arguing that paying interest on reserves was likely culprit and others blamed capital requirements. Everyone had their own story.

¹⁶ Many authors embarked on related projects regarding the transmission of monetary policy through banks, creating a thriving area: For example, [Piazzesi and Schneider \(2018\)](#) link settlement payments and securities prices, [Drechsler, Savov and Schnabl \(2017\)](#) competition in deposit markets, [De Fiore, Hoerova and Uhlig \(2018\)](#) think of collateral, [d'Avernas and Vandeweyer \(2024\)](#) connect with the bond market, [Ríos-Rull, Ritto, Takamura and Terajima \(2024\)](#) with default and others blend New Keynesian features ([Arce, Nuño, Thaler and Thomas 2019](#); [Piazzesi, Rogers and Schneider 2019](#)).

Where do convenience yields come from and how do they connect to monetary policy?

Papers: [Bianchi and Bigio \(2026, forthcoming, JET\)](#); [Bigio, D’Erasmus and Yezzer \(2025c, draft\)](#); [Bigio, Linzert, Mendo, Thaler and Schumacher \(2024b, in production\)](#).

When we started working on [Bianchi and Bigio](#), the penalties $\chi^\pm$ came from banks that bargained once in an over-the-counter market. That was fine for the theory but unappealing: the interbank rate was constant. I was again lucky to attend the right seminar at the right time: Ricardo Lagos was presenting [Afonso and Lagos \(2015\)](#) at the Philly Fed. That paper had the right ingredients: dynamic search that allows rates—the terms of trade—to vary depending on the quantities. Yet, [Afonso and Lagos \(2015\)](#) was challenging to work with because one needs to keep track of distributions of bank positions. If there were a way to kill that, maybe we could embed a block like that into [Bianchi and Bigio](#). If we could eliminate that state variable, perhaps we could embed the block in [Bianchi and Bigio](#).

I was lucky again that my three mentors at UCLA had a joint paper that contains the “trick” we needed ([Atkeson, Eisefeldt and Weill, 2015](#)). What began as a small appendix became [Bianchi and Bigio \(2026\)](#). In that paper, we extend our earlier static formulation into a general portfolio theory in which asset trades must settle, and settlement runs through over-the-counter frictions. The theory delivers convenience yields, the function $\chi^\pm$, trading probabilities, and the average interbank rate in closed form. All you need to know is the initial market tightness and the terminal tightness after a trading session is over, $\bar{\theta}$ —this, in turn, is a solution to an ODE given by a matching function. The convenience-yield coefficients are the corridor spread scaled by auxiliary weights, $\bar{\chi}^+ = (R^w - R^m) \zeta^+(\theta)$ and $\bar{\chi}^- = (R^w - R^m) \zeta^-(\theta)$, where

$$\zeta^+(\theta) \equiv \left(\frac{\bar{\theta}}{\theta}\right)^\eta \left(\frac{\theta^\eta \bar{\theta}^{1-\eta} - \theta}{\bar{\theta} - 1}\right) \quad \text{and} \quad \zeta^-(\theta) \equiv \left(\frac{\bar{\theta}}{\theta}\right)^\eta \left(\frac{\theta^\eta \bar{\theta}^{1-\eta} - 1}{\bar{\theta} - 1}\right),$$

and R^m and R^w are the outside options for borrowers and lenders.

The formula works for any matching function. But the matching function determines the speed at which the short side of the market is exhausted, and the paper compares what happens as the matching function moves from Cobb–Douglas to Leontief—two extremes for which the theory also delivers a closed-form solution for $\bar{\theta}$. The Cobb–Douglas solution is particularly beautiful because it delivers cases where one side of the market is exhausted in finite time.

A takeaway: the kink between $\bar{\chi}^-$ and $\bar{\chi}^+$ makes investors behave as if they were risk averse even under risk-neutral preferences: assets whose returns generate volatile settlement needs command higher convenience yields. I think this is a promising avenue to think about monetary policy and how it affects asset prices, by impacting liquidity.

I have been putting this machinery to work on two policy-related papers: [Bigio et al. \(2025c\)](#), joint with Pablo D’Erasmus and Luis Yezzer, estimates the demand for reserves in the US across monetary-policy implementation regimes. In [Bigio et al. \(2024b\)](#), we construct comparative statics to understand the parameters that govern the effects of open-market operations.

How does dollar liquidity affect exchange rates?

Papers: [Bianchi, Bigio and Engel \(2025, R&R, Restud\)](#); [Bigio, Castillo and Han \(2025d, draft\)](#).

There's something magical about the international payments system. You can be a Korean importer, travel to France, buy wine, and bring it back to Korea. Here's the magic: the French exporter has no idea who the Korean importer is, or which bank gave him a credit line or a checkbook. All the French exporter needs to know is that his French bank accepts funds from Korean banks. If you read Stigum ([Stigum and Crescenzi, 2007](#)), it's clear what is going on: the Korean bank creates dollar deposits which circulate around the globe. Of course, that needs a settlement instrument, dollar reserves. If a bank is short of reserves, it has to tap the LIBOR market. What this observation means for the dollar and its counterpart exchange rates is the starting point of [Bianchi et al. \(2025\)](#).

This is what we do. Rather than thinking of exchange-rate determination as a multi-country problem, we think of the world as a whole, as a closed economy with two central banks. The idea is that citizens of the world can borrow from and lend to banks anywhere in the world—hence, we strip down everything to a two-currency version of [Bianchi and Bigio \(2022\)](#). The key equation is the dollar liquidity premium:

$$\underbrace{(1+i) - (1+i^*) \frac{\mathbb{E}[e']}{e}}_{DLP} = \mathcal{LP}^* - \mathcal{LP},$$

where $*$ denotes a dollar variable, i an interest rate, e the exchange rate and $\mathcal{LP}$ the liquidity premium encountered above. Because international payments are disproportionately settled in dollars, the liquidity premium of the dollar will exceed that of the euro, commanding a currency convenience yield that depends on policies and, of course, the liquidity ratios μ by currency. The premia are exactly the convenience-yield formulas of the settlement theory I describe above. Deviations from covered and uncovered interest parity are manifestations of this premium and can be amplified by risk premia. When the dollar's liquidity premium rises because of dollar scarcity, then, holding rates fixed, the exchange rate must adjust—together with its expected future value, a function of the process that drives shocks.

The paper takes these insights and does three things. First, it finds evidence of a correlation between μ and the exchange rate. I remember the excitement of my co-author, Charles Engel, telling us how after decades of looking into the drivers of exchange rates, this was the first instances where he found quantity variable appeared significantly in an exchange-rate regression. Second, we derive analytically the implied predictions of how the exchange rate moves in relation to monetary-policy shocks and shocks to the settlement risk in dollars. Finally, we conduct an inference analysis showing that periods of dollar appreciation and CIP deviations align well with the theory in terms of the movements in funding flows.

We began thinking about this paper around 2014—when [Gabaix and Maggiori \(2015\)](#) was already being presented and reviving the literature on financial intermediaries and exchange rates. Several others connected this force with classic puzzles in international economics—prominently [Itskhoki and Mukhin \(2021\)](#). Let me tell

you what's totally different in our work. In our work, the composition of assets *within* the same currency matters, not only the net position in it. Two portfolios with identical dollar exposure but different mixes of reserves and deposits will deliver completely different exchange rates and CIP deviations. Settlement frictions therefore give a role to swap lines, sterilized interventions, and repo facilities; it also gives a role to movements *within* the capital account—which net out in conventional accounting. Think about what a capital flight is: is it that investors pull physical capital out of a country or that the country experiences an export boom to pay for bonds? Or is it that investors sell assets, get deposits and drain bank reserves from a country? In our view, it is the shortage of dollars that impacts the exchange rate and the financial sector's willingness to lend. Current-account movements alone cannot reveal this mechanism; the key is to look at the composition of gross flows.

Our paper also has a small-open-economy spin-off: In [Bigio et al. \(2025d\)](#), joint with Paul Castillo (Central Reserve Bank of Peru's Managing Director and former head of operations) and Seungyub Han, a former student of UCLA, we take the framework to a small open economy, tracing how the inflow and outflow of dollar liquidity shape credit conditions and the exchange rate. Peru is an excellent testing ground because we observe detailed short-term assets and liabilities by currency.

Why not satiate banks with liquidity? What is the optimal size of the central bank's balance sheet?

Papers: [Bigio and Sannikov \(2024\)](#), *R&R, Journal of Finance*.

The most important insight of [Bianchi and Bigio \(2022\)](#) is that central banks have more tools than conceived by conventional wisdom: central banks can control spreads in addition to rates. But here's the question that also haunted me seminars: nice theory, but why would any central bank want to control spreads? I mean, why not set spreads to zero by satiating banks with reserves? After all, spreads are distortions and distortions are, in principle, not good. This question is coming back today as central banks are re-designing their operational targets.

Thanks to a year-long fellowship at Princeton in 2015, I met Yuliy Sannikov, now at Stanford GSB. In [Bigio and Sannikov \(2024\)](#) we began thinking of the normative implications of this insight in an incomplete-markets monetary economy. On the one hand, interest-rate targets grant control over inflation, but the supply of reserves grants control over credit spreads. By adequately choosing open-market operations (OMO) and setting policy rates, central banks can target inflation and the volume of credit in the economy. The central bank can operate in three regimes: a corridor system, in which it has control over interest rates and credit spreads; a satiation regime, where the transmission channel is the New Keynesian model; and a liquidity-trap regime, where credit spreads are zero and OMO are neutral, but reductions in interest rates beyond a negative threshold are recessionary.

A message that emerges is that the central bank should operate with a small balance sheet during booms but expand its balance sheet, operating in a floor system, during busts—and it should never drop rates past the negative point where they trigger a liquidity trap. This recommendation results from a complex policy

tradeoff. The smaller the central bank's balance sheet, the higher the credit spreads. Credit spreads, altered through OMO, have direct effects through the credit channel. By affecting credit, spreads can enhance the effects of the standard instrument and increase the stabilization power of monetary policy. By tightening credit, monetary policy can also mitigate the impact of aggregate shocks—a standalone macro-prudential effect. The stabilization power and the macro-prudential benefits contribute to overall macro insurance. The tradeoff is that increasing credit spreads reduces insurance against idiosyncratic shocks.

That paper went through many incarnations, starting from a different source of real effects of monetary policy all the way to finally incorporating elements of the New Keynesian model. The paper also made a theoretical contribution: a technical challenge in evaluating macro-prudential policies in heterogeneous-agent economies is that the welfare case for macroeconomic insurance depends on what agents believe about the shock prior to its arrival. The common device of an unanticipated “MIT” shock is ill-suited for normative analysis because it does not allow households to anticipate risk and self-insure accordingly. Solving the full rational-expectations equilibrium with distributions as state variables requires an approximation. Our risk-adjusted steady state (RSS) approach does not. It addresses this by letting a single aggregate shock event arrive with Poisson intensity. Before arrival, the economy rests at a stationary point where agents anticipate the risk. Upon arrival, the economy follows a perfect-foresight transition. Computationally, our approach just adds one more date to the path of an MIT shock. Beyond consistency, the approach allows for a clean welfare decomposition.¹⁷

¹⁷ This is simple to work with and a good resource. I hope other catch up on this.

Why do bank liabilities circulate, i.e., why are these money?

Papers: Bigio et al. (2024a, draft); Bigio, Weill and Zuniga (2026b, draft).

As noted above, many banking models neither explain why banks exist nor distinguish them from other firms. As a monetarist, I believe that banks are unique because their liabilities are money, and that monetary services matter.¹⁸ In practice, when a bank makes a loan to a business owner, the bank acquires an asset, and, in exchange, the borrower receives a deposit. Deposits are different from any other financial liabilities because they circulate: borrowers receive deposits that they later use to pay workers, and workers make further purchases, and so on.

¹⁸ Money and banking was historically taught as a unified course. Have you ever asked why?

As happens with the international payment system, we do not have to worry about which bank issued what liability when we accept payments. Banks settle that for us. We do not know what happens behind the scenes, we just know in which bank our deposit account is.¹⁹ These observations suggest that the creation of circulating liabilities has significant social value. Yet, when we turn to classic models in banking (Diamond and Dybvig, 1983; Diamond, 1984; Williamson, 1987; Boyd and Prescott, 1986), these theories leave aside almost entirely the notion that bank liabilities are the primary medium of exchange.

¹⁹ The topic of joint liability merits its own obsession.

In ongoing work with my UCLA colleague Pierre-Olivier Weill and our former student Diego Zúñiga, now at the Bank of Canada (Bigio et al., 2024a), we formalize what is distinctive about banks as providers of circulating deposits. We start from the premise that many assets in the economy cannot circulate because of

asymmetric information, but others could, at least in principle, if all parties were equally informed.²⁰ We think of banks as financial institutions that purchase these illiquid assets in exchange for their deposits, which do circulate. Since illiquid assets provide the backing for the value of deposits, deposits could potentially also inherit the liquidity problem. In our setting, banks structure their liabilities, stripping out the information content to guarantee their circulation, an idea due to [Gorton and Pennacchi \(1990\)](#). Our contribution is to do so in a competitive general equilibrium environment.

The core of the theory is one condition. Take any security that pays x_ℓ in one state and x_h in the other, with corresponding probabilities π , and hand it to a producer who knows the state and creates added value ρ from trading the security. The security circulates as a means of payment if and only if its payoff is not too tilted in either state:

$$\underbrace{\psi_\ell \equiv \frac{1 - \rho\pi_\ell}{\rho\pi_h}}_{<1} \leq \frac{x_h}{x_\ell} \leq \underbrace{\psi_h \equiv \frac{\rho\pi_\ell}{1 - \rho\pi_h}}_{>1}.$$

Securities trade at their pooling price provided that their payoffs fall in that cone. Outside it, trade collapses. As a result of competition, banks bundle every asset on the left side of the balance sheet and tranche their payoffs to support the largest circulating deposit. The other tranche becomes bank equity.

Three implications. One: narrow banks are a bad idea—swapping liquid assets for equally liquid liabilities adds no value. Two: liquid and illiquid assets are complements in liquidity transformation; a liquid asset is worth more inside a bundle and vice versa. Three: deposits should not be riskless.

How are banks constrained, and when does interest-rate risk matter?

Papers: [Begenau, Bigio, Majerovitz and Vieyra \(2026, Restud\)](#); [Abad, Bigio, Garcia-Villegas, Marbet and Nuno \(2025, draft\)](#).

As the opening emphasized, one rarely knows where economic exploration will lead. When trying to calibrate the intermediary models in [Bigio and d’Avernas](#) and [Bianchi and Bigio](#), the goal was to produce dynamics for bank equity that would replicate the patterns of the data. I did not find an obvious way to do so. In 2014, I met Julianne Begenau—she presented her job-market paper [Begenau \(2020\)](#)—and confronted the same challenge: when targeting the dynamics of book equity, models deliver little action. Targeting market values produces counterfactual procyclical behavior for leverage.²¹

It was clear that a theory that relies on constraints on market or book values alone is not enough. We began to work with two now-former students, Jeremy Majerovitz, a rising macro-development star at Notre Dame, and Matias Vieyra. The starting point in [Begenau et al. \(2026\)](#) is a set of facts about bank Tobin’s Q that provide a unifying concept to think about what actually constrains banks:

1. The time-series behavior of Tobin’s Q reflects that bank book and market values diverge during crises.
2. Market equity captures information not contained in the book because Tobin’s Q has predictive power.

²⁰ This view is also rooted in the UCLA tradition—in the views of Alchian, in particular.

²¹ Of course, other scholars were finding similar difficulties (see [Adrian, Colla and Shin, 2012](#); [He et al., 2010](#); [Adrian, Etula and Muir, 2014](#)) and our initial purely empirical paper morphed into a theory.

3. The cross-sectional dispersion of Tobin's Q increased during a banking crisis.
4. Individually, Tobin's Q and market-leverage are slowly mean-reverting after a shock to net worth.

We contend that the most prominent models with intermediaries cannot jointly deliver these facts (Gertler and Kiyotaki, 2010; He et al., 2010; Brunnermeier and Sannikov, 2014), at least not in their original form.

In the paper, we argue that both market and book equity measures matter and present a model with delayed accounting that matches all facts: banks are constrained by book equity through regulation. The core idea is that banks have the ability to postpone recognizing their losses in their books; thereby, when a crisis hits, market values erode, market leverage fans out, but book leverage takes time to respond. Part of the paper also analyzes the advantages of faster accounting of bank losses.

More recently in Abad et al. (2025), joint work with Jorge Abad, Salomon Garcia-Villegas, Joel Marbet, and Galo Nuño at the Bank of Spain, we study why regulatory constraints are essential for understanding the pass-through of policy rates to loan rates in fixed- and variable-rate systems. That paper shows that a distinct pass-through exists between fixed- and variable-rate systems if regulatory constraints bind for many banks. Because the relevance of these constraints is a numbers question, this is a quantitative heterogeneous banking model aimed at replicating the pass-through of different systems pass-through across systems as a function of the distribution of bank equity.

How does liquidity shape the optimal issuance of government debt?

Papers: Bigio, Nuno and Passadore (2023, JPE: Macro); Bigio and Nuno (2025, R&R, AEJ: Macro).

A final example of cross-paper spillovers is Bigio et al. (2023), joint work with Juan Passadore of the IMF and Galo Nuño of the Bank of Spain. This paper studies the optimal maturity of public debt. On the surface, this topic has nothing to do with the agenda above. It came out of a conversation with Juan at Einaudi, who was thinking about optimal maturity issuance, and with Galo, who was working on an impressive paper about controlling distributions (Nuno and Moll, 2018).

Our paper makes two conceptual innovations relative to public finance papers. First, we highlight the role of liquidity frictions, the notion that auctioned quantities in the primary bond market impact prices, depending on the maturity of each auction. What could give rise to that price impact? These are the same frictions that inspire the interbank market of Bianchi and Bigio (2026): if intermediaries enter auctions to distribute bonds in an OTC secondary market, greater issuances signify a greater carry cost. The second innovation is introducing a distribution of bonds of different maturity and dynamically optimizing over it. The density of bonds replaces the density of people in Nuno and Moll (2018)—and thanks to my work in Bigio and Sannikov (2024), the connection was clear. In Bigio et al. we derive a simple principle that guides the optimal issuance of bonds of a given maturity:

$$\frac{\text{issuance at maturity } \tau}{\text{GDP}} = \frac{\text{value gap at maturity } \tau}{\text{liquidity coefficient at maturity } \tau}.$$

The principle states that in a given period, the issuance of a bond of a given maturity (relative to GDP) is given by the value gap divided by the liquidity coefficient. The value gap is the difference between the market price, or asset-pricing kernel, and the internal valuation of a bond, an internal pricing kernel, as a percentage of the price. In this model, pricing kernels are not equalized because of the liquidity frictions. Issuances map into an evolving distribution of debt that feeds back as an internal pricing kernel. We estimate liquidity costs and calibrate the model to evaluate Spain's debt management practices.

In the sequel, [Bigio and Nuno \(2025\)](#), we employ the risk-adjusted steady-state (RSS) approach to decompose the forces behind optimal debt maturity, splitting the role of risk from that of liquidity.

Seemingly unrelated work—but not entirely

Papers: [Bigio, Silva and Zilberman \(2026a, forthcoming, JFE\)](#); [Bigio and Ishide \(2025, draft\)](#); [Bigio, Caramp and Silva \(2025b, R&R, JPE: Macro\)](#).

Here are some further unexpected connections. Because of my work, I think monetary policy affects spreads; and I am concerned that, inadvertently, it may inadvertently inflate asset prices. I wanted to think about whether ample liquidity fuels asset prices and speculation. That was the original motivation in [Bigio et al. \(2026a\)](#), joint with Dejanir Silva and Eduardo Zilberman. We initially envisioned a Harrison–Kreps model fueled by monetary policy with business-cycle effects. But, we soon ran into problems: how to think of asset prices having real effects, how to think of a theory of the firm under incomplete markets, what beliefs to impose, all questions that needed a benchmark model. We decided to go back to basics and build a benchmark business-cycle model where investors hold heterogeneous beliefs and asset prices reflect different views. Surprisingly, hiring in advance—an operational leverage channel—coupled with heterogeneous investors is enough to obtain a reasonable workhorse that matches asset prices and business cycles.

Earlier in this program, I was reluctant to work on the New Keynesian model because I thought many other dimensions were being neglected. Several rounds of revisions of [Bigio and Sannikov \(2024\)](#) eventually led us close to the New Keynesian block. I have since become fascinated with its elegance and few models give policymakers direct control over real interest rates; [Bianchi and Bigio \(2022\)](#) does, although prices may adjust too quickly there. As much as I am fascinated by its elegance, I'm concerned about its lack of robust conclusions.²² An earlier version of [Bigio and Sannikov \(2024\)](#) introduced a New Keynesian block that operates through the extensive margin—unemployment—as opposed to the intensive margin—hours. We soon found that its predictions are not robust to that switch and in [Bigio and Ishide \(2025\)](#), joint with my former student Akira Ishide, we formalize this point. In [Bigio et al. \(2025b\)](#), joint with Nicolas Caramp and Dejanir Silva, we look at the predictions of the New Keynesian model when the level of debt signals potential future event where inflation is needed to make debt sustainable. This paper is another example where a modest change in the environment reverses the model's standard predictions.

²² Contrast its fragility to that of the neoclassical growth model.

So what is the takeaway from all this?

Several conclusions connect these projects.

1. **Liquidity matters most where firms trade with each other.** Financial frictions on investment cannot produce deep recessions—investment is simply too small relative to the capital stock. Frictions on operating finance—payroll, intermediaries—are much more powerful (Bigio and La'O).
2. **Payment-chain disruptions can slow down supply chains.** Firms waste time as they wait to be paid. As they do so, they delay other firms' production. In crises, policy should focus on accelerating payments—including government payments (Bigio et al.).
3. **Banks create liabilities that circulate.** That is an enormous social benefit. Regulation that treats banks as mere intermediaries misses their monetary function—capital rules, safe-asset provision, and competition policy all have a monetary dimension (Bigio et al.).
4. **Central banks monopolize the settlement instrument, and that monopoly is a policy tool.** Deposit circulation requires a settlement instrument, and central banks monopolize its supply. By tilting the trade-off between liquidity risk and intermediation profits, the supply of reserves steers credit spreads and the volume of lending (Bianchi and Bigio).
5. **Central banks control more instruments than conventionally thought.** This is an implication of the previous point.
6. **Satiation is not free: the optimal balance sheet is state-contingent.** Credit spreads are distortions, but distortions can have a macro-prudential value. The central bank should run a lean balance sheet in booms, expand into a floor system in busts, and never push rates past negative rates (Bigio and Sannikov).
7. **The dollar's role in settlement gives policy a lever over convenience yields.** Because international payments settle disproportionately in dollars, the composition of positions *within* the capital account—not only net exposures—moves exchange rates. That is what gives swap lines and repo facilities their bite, and why capital flight is better read as a scramble for settlement instruments than as an exodus from a country's assets (Bianchi et al.).
8. **Banks can circumvent regulation by delaying their losses.** Book and market equity diverge precisely in crises, so supervisors should weigh both (Begenau et al.).
9. **Repos are the market's answer to private information—but an imperfect one.** Repurchase contracts keep assets from sitting idle under adverse selection, yet competition induces cream-skimming relative to a planner, so a design margin remains for policy (Bigio and Shi).

For students, the lessons and technical tricks are scattered through the essay.

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PAPERS & WORK IN PROGRESS

The table below lists the papers produced or published after graduate school. Some are not included in this essay.

Paper	Coauthors	Status
Liquidity and Asymmetric Information		
Endogenous Liquidity and the Business Cycle	—	Published at AER 2015
Financial Frictions in Production Networks	La’O	Published at QJE 2020
Liquidity Shocks and the Business Cycle	Schneider	Published at EER 2017
Repurchase Options in the Market for Lemons	Shi	Published at Restud 2025
A Theory of Payment-Chain Crises	Mendez, Van Patten	Draft, 2025
Money and Banking		
Financial Risk Capacity	d’Avernas	Published at AEJ: Macro 2021
Liquidity Management and Monetary Policy	Bianchi	Published at Econometrica 2022
Portfolio Choice and Settlement Frictions	Bianchi	Forthcoming at JET
A Monetary Theory of Bank Balance Sheets	Weill, Zuniga	Draft, 2024
Competitive Bank Liability Design	Weill, Zuniga	Draft, 2026
A Q-Theory of Banks	Begenau, Majerovitz, Vieyra	Published at Restud 2026
Heterogeneous Beliefs, Asset Prices, and Business Cycles	Silva, Zilberman	Forthcoming at JFE
The Heterogeneous Bank Lending Channel of Monetary Policy	Abad, Garcia-Villegas, Marbet, Nuño	Draft, 2025
Monetary Policy Implementation Regimes and Reserve Demand	D’Erasmus, Yezpe	Draft, 2025
A Comparative-Statics Approach to Central Bank Balance Sheets	Linzert, Mendo, Schumacher, Thaler	In production
Money, Credit, and the New Keynesian Model		
A Model of Credit, Money, Interest, and Prices	Sannikov	R&R at Journal of Finance
New Keynesian Economics through the Extensive Margin	Ishide	Draft, 2025
Sticky Inflation	Caramp, Silva	R&R at JPE: Macro
Transfers vs. Credit Policy	Zhang, Zilberman	R&R at EER
International Liquidity		
Scrambling for Dollars	Bianchi, Engel	R&R at Restud
Dollar Liquidity Flows in Small-Open Economies	Castillo, Han	Draft, 2026
Departures		
Debt-Maturity Management with Liquidity Costs	Nuño, Passadore	Published at JPE: Macro 2023
Decomposing Optimal Debt Maturity	Nuño	R&R at AEJ: Macro
Carbon Pricing and Inequality	Känzig, Sanchez, Walsh	Forthcoming at EJ